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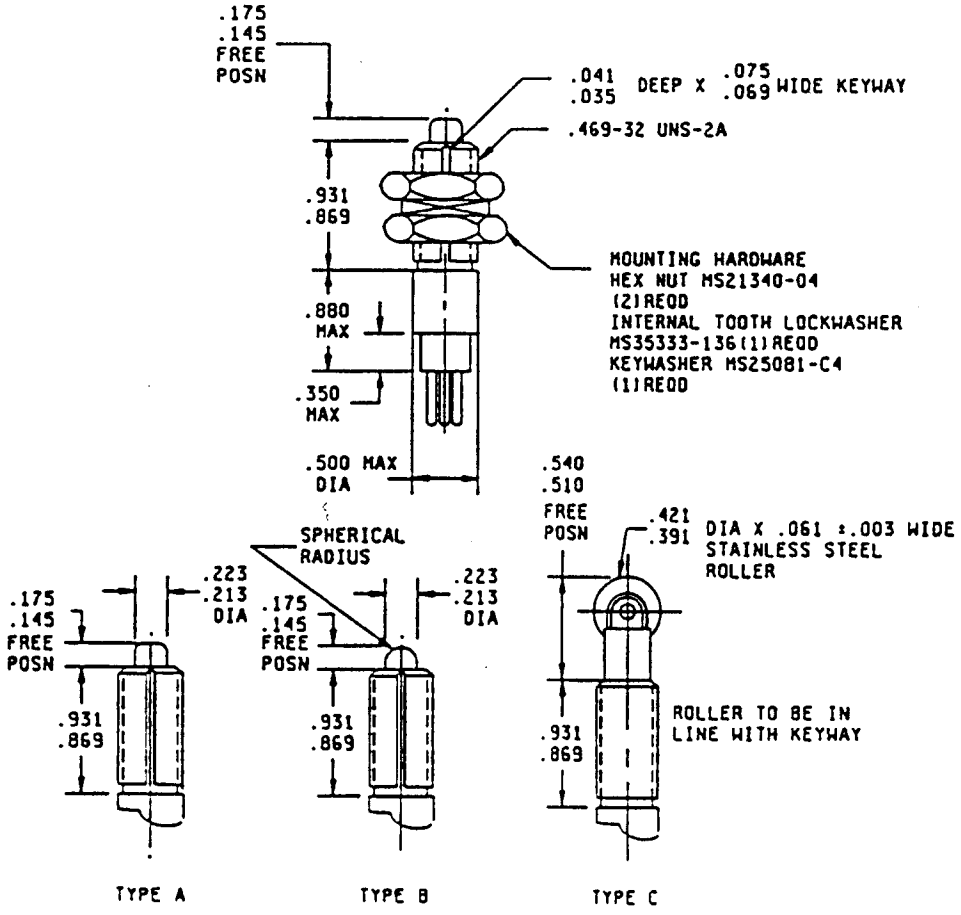
INCH-POUND
MIL-PRF-8805/74J
12 May 1995
SUPERSEDING
MIL-S-8805/74H
27 December 1991

PERFORMANCE SPECIFICATION SHEET

SWITCHES, SENSITIVE, SPDT, PLUNGER OR ROLLER, 7 AMPERES AND 4 AMPERES, HERMETICALLY SEALED CONTACTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-8805.

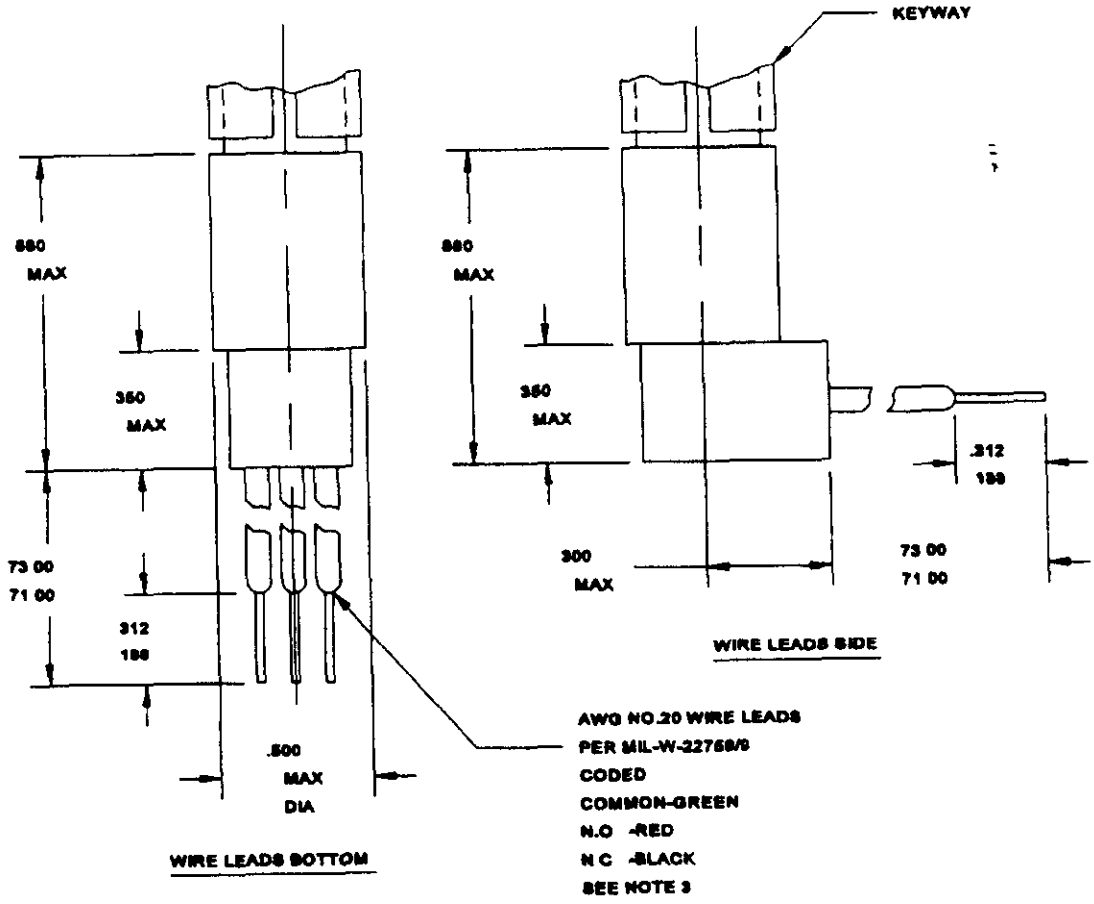


Ⓙ FIGURE 1. Dimensions and configurations.

Ⓙ denotes changes

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inches	mm	inches	mm	inches	mm	inches	mm
035	0 89	188	4 78	391	9 93	931	23 65
041	1 04	213	5 41	421	10 69	71 00	1803 4
047	1 19	223	5 66	469	11 91	73 00	1854 2
069	1 75	300	7 62	510	12 95		
075	1 91	312	7 92	540	13 72		
145	3 68	350	8 89	869	22 07		
175	4 45	385	9 78	880	22 35		

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only
3 Exact shape of switch optional within dimensions specified
(J) 4 Alternative base metals and protective finishes, as approved by the qualifying activity, may be utilized for lock washer Dimensions shall be in accordance with the referenced hardware

FIGURE 1 Dimensions and configurations Continued

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REQUIREMENTS

Dimensions and configurations See figure 1

(J) Enclosure design 5 (hermetic) Basic switch is hermetically sealed M8805/68-008, 022 listed on QPL 8805 Plunger seal as specified in table I

Temperature characteristic 2 (65°C to +125°C)

Shock type M (100 G, method 213, test condition I, MIL-STD 202)

Sinusoidal vibration grade 3 (10 to 3,000 Hz swept sinusoidal)

Operating characteristics See table II

Contact resistance 040 ohm plus 011 ohm per lead per foot

  After electrical endurance In accordance with MIL-S-8805, plus 011 ohm per lead per foot

Dielectric withstanding voltage

  Sea level and 70,000 feet 1,050 V rms (terminal to case), 800 V rms (terminal to terminal), 500 V rms after electrical life

Mechanical endurance 50,000 cycles

Electrical endurance 25,000 cycles

Intermediate current 25,000 cycles

Low level circuit 25,000 cycles

Electrical ratings See table III

Short circuit 20 times rated resistive load (MS3320-7 1/2 used for circuit protection)

Weight 1 0 ounce plus 0 1 ounce per lead per foot

Qualification

  Group submission See table IV

Group A inspection, seal test Documented inspection of the basic switch may be used to satisfy the seal test requirement

Military Part or Identifying Number (PIN) See tables I and V

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(J) TABLE I PIN s and configuration

PIN 1/	Load rating	Plunger seal	Type of actuating mechanism	Terminals
M8805/74 023	7 Amps	None	A	Wire leads bottom
M8805/74-024	7 Amps		C	Wire leads bottom
M8805/74-025	7 Amps		A	Wire leads side
M8805/74 026	7 Amps		C	Wire leads side
M8805/74-027	7 Amps	Resilient seal	A	Wire leads bottom
M8805/74 028	7 Amps		C	Wire leads bottom
M8805/74-029	7 Amps		A	Wire leads side
M8805/74-030	7 Amps		C	Wire leads - side
M8805/74 031	7 Amps	None	D	Wire leads bottom
M8805/74 032	7 Amps		D	Wire leads side
M8805/74-033	7 Amps	Resilient seal	D	Wire leads bottom
M8805/74-034	7 Amps		D	Wire leads side
M8805/74-035	4 Amps	None	A	Wire leads - bottom
M8805/74-036	4 Amps		C	Wire leads - bottom
M8805/74-037	4 Amps		A	Wire leads side
M8805/74 038	4 Amps		C	Wire leads side
M8805/74-039	4 Amps	Resilient seal	A	Wire leads bottom
M8805/74 040	4 Amps		C	Wire leads bottom
M8805/74-041	4 Amps		A	Wire leads side
M8805/74 042	4 Amps		C	Wire leads side
M8805/74-043	4 Amps	None	D	Wire leads - bottom
M8805/74-044	4 Amps		D	Wire leads side
M8805/74-045	4 Amps	Resilient seal	D	Wire leads bottom
M8805/74 046	4 Amps		D	Wire leads - side

1/ Part numbers M8805/74-001 through 022 have been canceled See table V for substitution information

TABLE II Operating characteristics

Plunger seal	Actuating force (maximum)	Releasing force (minimum)	Movement differential (maximum)	Pretravel inches (maximum)	Overtravel (minimum)
Unsealed	30 ounces 1/	4 ounces 1/	005 inch (0 13 mm)	010 (0 25 mm)	125 inch (3 18 mm)
Resilient	7 0 pounds	1 5 pounds	005 inch (0 13 mm)	020 (0 51 mm)	125 inch (3 18 mm)

1/ At sea level and 70,000 feet

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(J) TABLE III Electrical ratings

Load	Sea level and 70,000 feet	
	28 V dc	
	(Amperes)	(Amperes)
Resistive	7	4
Inductive	1	1
Lamp	2	2

(J) TABLE IV Qualification

Inspection	Basic switch	Additional inspection	Extent of approval
Qualification inspection table of MIL S-8805, groups I through VI, inclusive	M8805/74-028, 040 (all sample units) (028 will qualify the 040 because of amp rating but not the reverse)	M8805/74-023 through 027, 029 throuth 039, 041, 042 through 046 (2 sample units each) Visual and mechanical inspection only	All

(J) TABLE V Substitution information for canceled PIN s 1/

Canceled PIN s M8805/74-	7 Amp load PIN's M8805/74-	4 Amp load PIN's M8805/74-
004	023	035
006	024	036
007	025	037
009	026	038
013	027	039
015	028	040
016	029	041
018	030	042
019	031	043
020	032	044
021	033	045
022	034	046

1/ The canceled PIN s are no longer acquirable The substitute parts are interchangeable in all respects except electrical ratings The inactive parts utilized M8805/68 001 basic switches rated at sea level and 70,000 feet as follows

28 V dc 7 A resistive, 3 A inductive, 2 A lamp load
115 V ac, 400 Hz 5 A resistive, 3 A inductive, 2 A lamp load

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CONCLUDING MATERIAL

Custodians
Army ER
Navy EC
Air Force 85

Review activities
Army AR AV, MI
Navy AS, MC, OS, SH
Air Force 19, 99

Preparing activity
DLA ES

(Project 5930 1594 05)*

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